





Monitoring of Stream Processing Engines Beyond the Cloud: an Overview

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Introduction

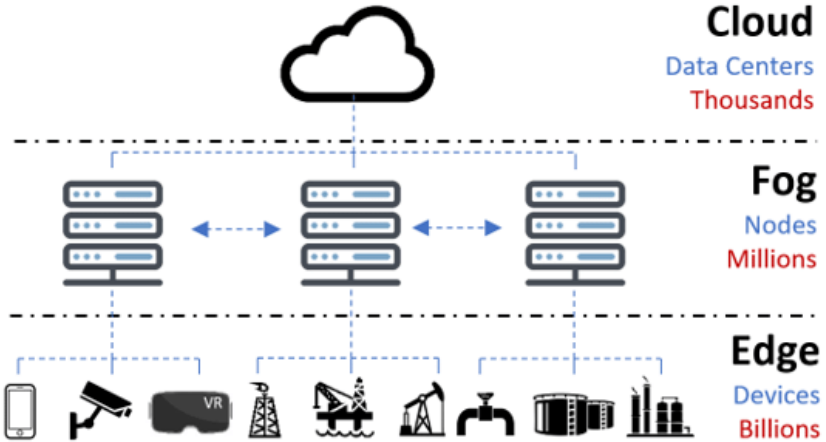
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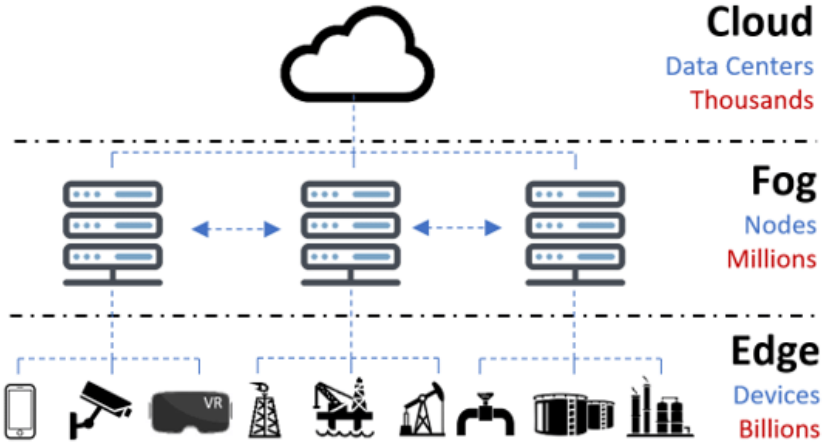
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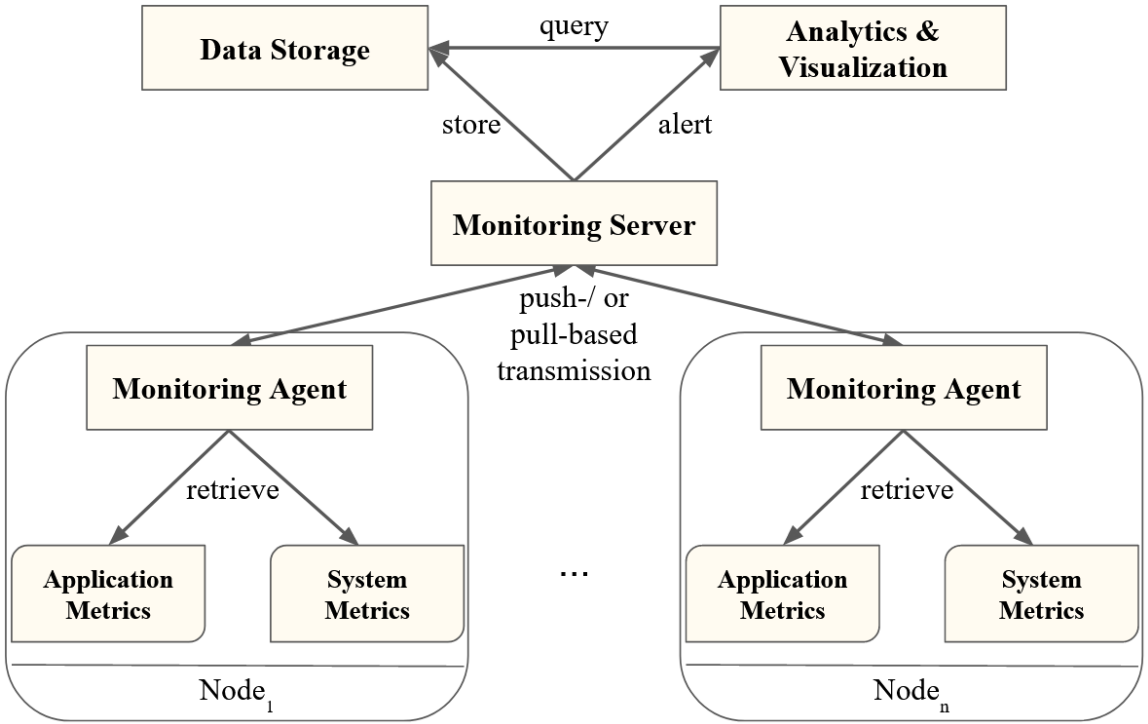
Contributions:

- Experimentally evaluate different monitoring scenarios in an IoT environment
- Highlight inefficiencies of existing approaches
- Define a set of functional requirements for a suitable monitoring solution

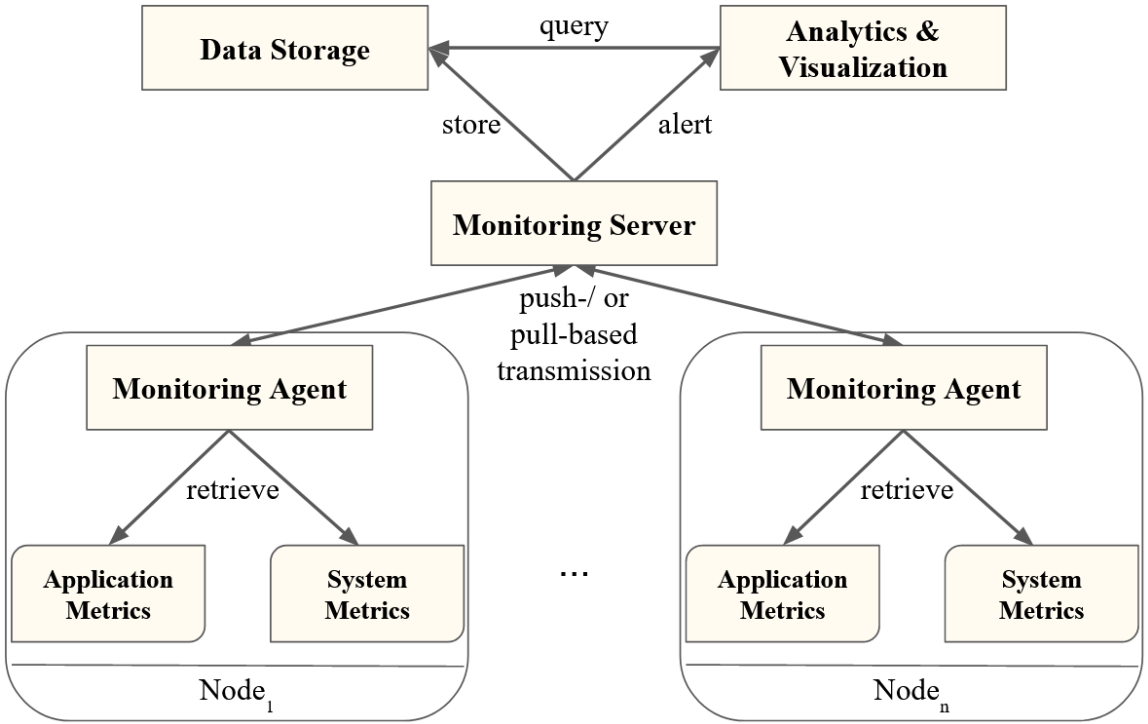


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General-purpose Monitoring



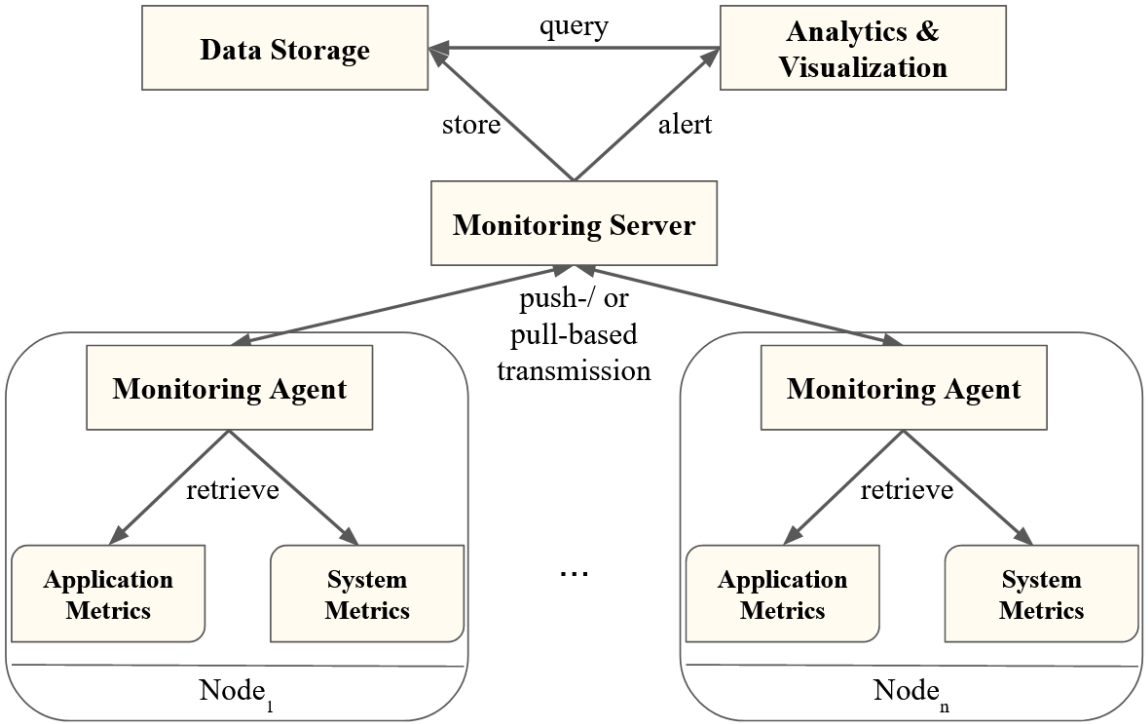
General-purpose Monitoring



Prometheus

Grafana

General-purpose Monitoring



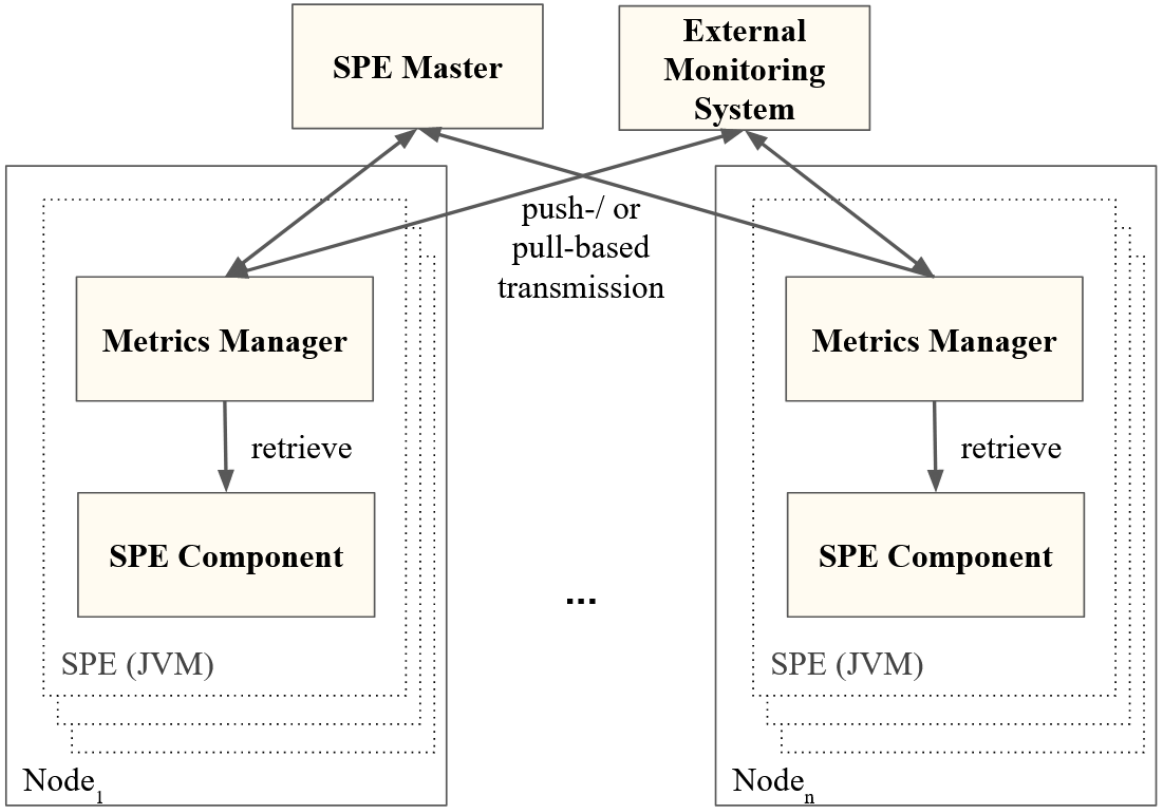
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Drawbacks:

- Dependency to an external system
- Internal monitoring requires crossing of system boundaries

Monitoring in Cloud-based SPEs



Apache Flink

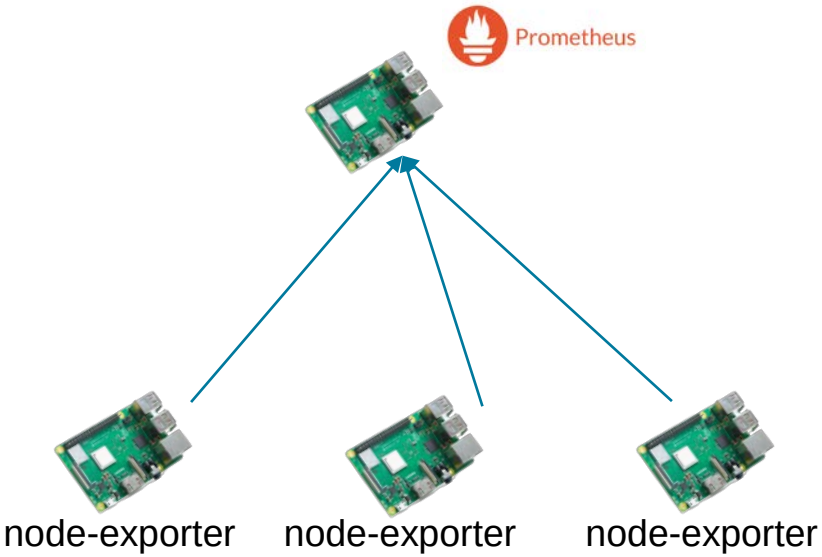
- Same approach of metric transmission as external monitoring systems
- Integration with external monitoring systems via APIs (e.g., REST)

Experimental Evaluation

- Representative monitoring solution:
  - Prometheus ecosystem

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- Setup:
  - Four RaspberryPi 4 Model B
  - 1.5 GHz 64-bit quad core ARM Cortex-A72
  - 4GB of LPDDR4-3200 SDRAM
  - RPIs are connected over an Ethernet switch
  - Energy supply via a central USB power hub



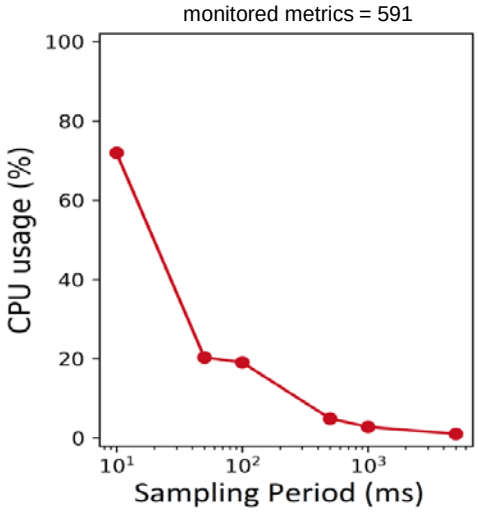
Experimental Design

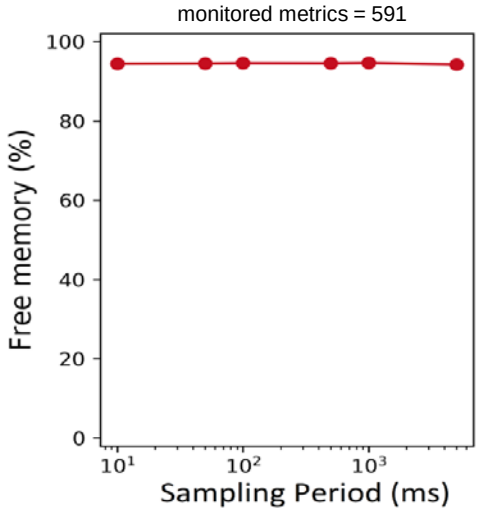
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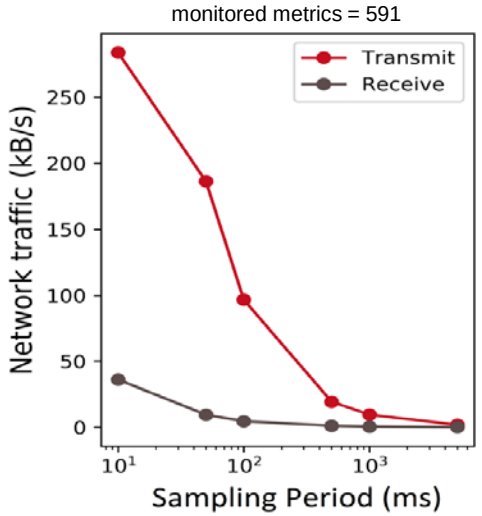
2) How does system load affect monitoring?

3) How do large numbers of metrics affect monitoring?

Resource Utilization







Small sampling periods limit available CPU resources and bandwidth

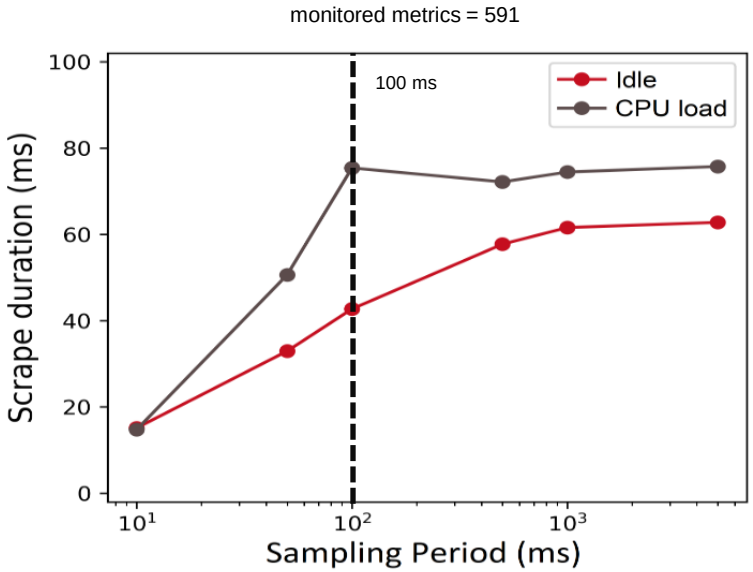
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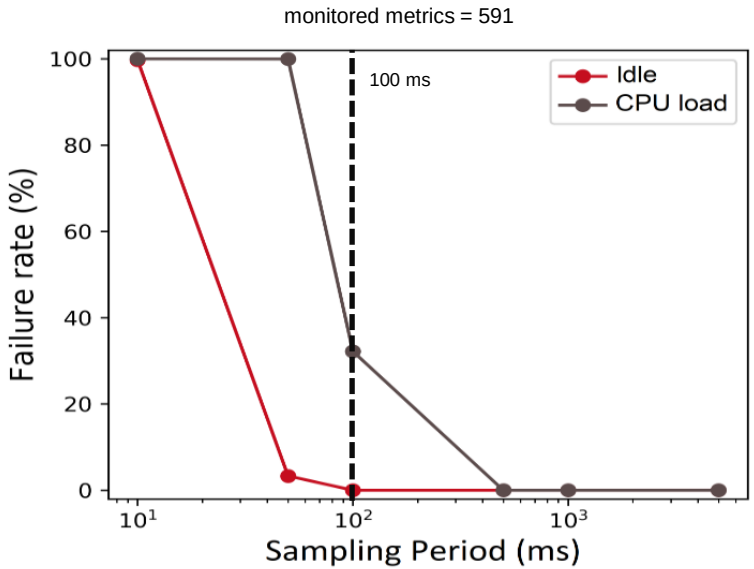
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Impact of CPU Load





System cannot handle low latency (< 100 ms) monitoring jobs

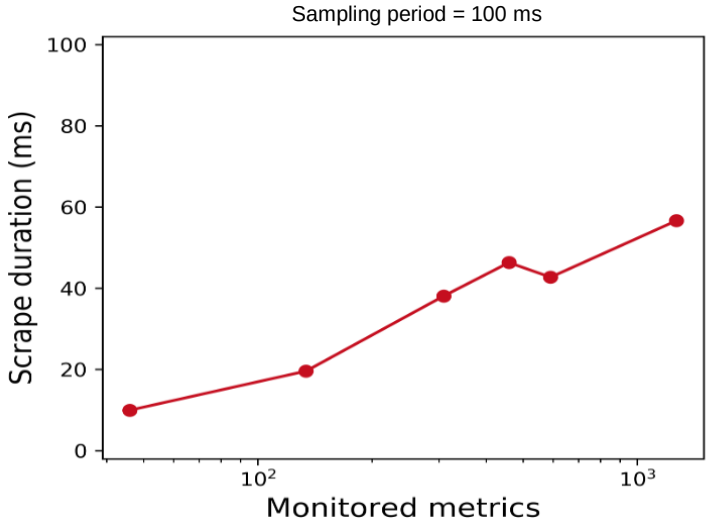
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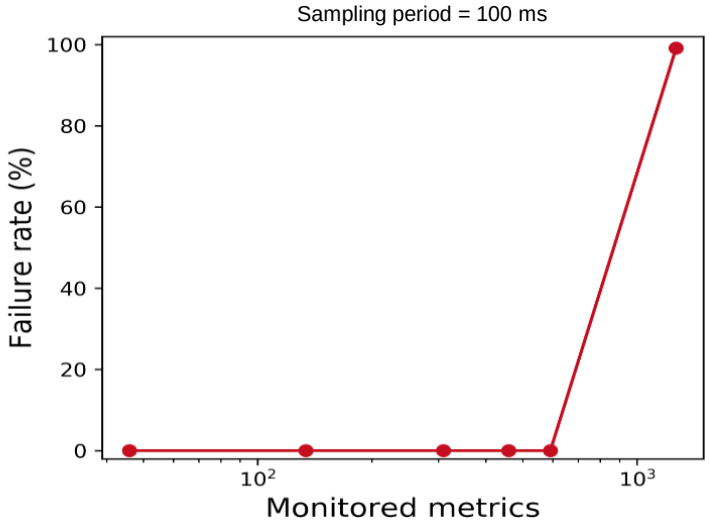
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Number of Metrics Experiment





Network limit reached at 1268 metrics

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- Performance Optimization and Scalability
  - Filtering of measured metrics
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- Handling Heterogeneity
  - Support of different OSs and devices
  - Standardized communication

Conclusion

- Current SPE monitoring solutions are tailored for cloud infrastructures and inefficient for IoT topologies

- We identify four classes of functional requirements that are needed to enable monitoring in the IoT:
  - Performance Optimization and Scalability
  - Handling Uncertainties
  - Permission and Access Control
  - Heterogeneity

